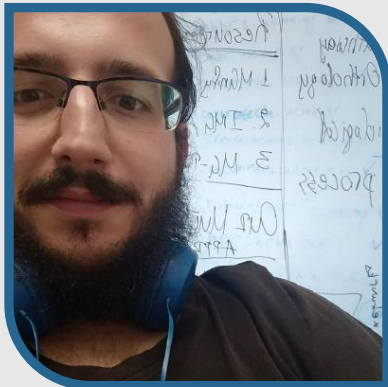




Zafeiropoulos Haris

Bioinformatics, Systems Biology, Microbial Ecology

-  April 24, 1991
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-  haris.zafr@gmail.com

About Me

I develop bioinformatics approaches to better understand microbial communities, their dynamics and microbial interactions. My main scientific focus is on microbial interactions—especially metabolic ones—and their evolutionary implications. I am particularly interested in how microbial communities are shaped by their environment and, in turn, how their structure feeds back to influence that environment. To address these questions, I use diverse approaches, spanning 'omics, network analysis, and metabolic modeling.

Social Network

-  [Github Profile](#)
-  [LinkedIn Profile](#)
-  [Google Scholar Profile](#)
-  [ResearchGate Profile](#)

Languages

-  Greek ● ● ● ● ●
-  English ● ● ● ● ●
-  French ● ● ● ● ●

Research projects - working experience

Current Employment status

- 2026 - ongoing **Predicting facultatives** Post-doc
Recently, I joined the Cluster of Excellence "Balance of the Microverse" and the "Mechanisms of microbial metabolic interactions (m3i)" lab, led by Dr. Zimmermann, at the Friedrich Schiller University of Jena. **My role** will be to investigate machine learning approaches for accurate prediction of facultative bacterial taxa based on their genomes. Moreover, I will further develop metabolic modeling based methods for microbial interactions inference.

Formal Employment Projects

- 2022 - 2026 **3D'omics** Post-doc
Aim of the project is to come up with technology and protocols to generate 3D omics landscapes, achieving reconstructions of intestinal host microbiota ecosystems. **My role** on the project is the analysis of microbial networks integrating spatial and metabolic data to predict species interactions and/or influence of metabolites on species distributions using co-occurrence network analysis and Genome-Scale Metabolic Modeling.
- 2022 - 2022 **A workflow for marine Genomic Observatories data analysis** Research associate
An EOSC-Life-supported project aimed at making large volumes of sequencing data produced by genomic observatories easier to interpret providing taxonomic inventories in a timely and non-technical format. **My role** was the main developer of the workflow, and the Principal Investigator in the second half of the project.
- 2018 - 2021 **PREGO: Process, environment, organism (PREGO)** PhD
PREGO is a systems-biology approach to elucidate ecosystem function at the microbial dimension. PREGO was the main project of my PhD and **my role** was to develop the data integration parts of the platform.
- 2019 - present **ELIXIR-GR** technical support
ELIXIR-GR is the Greek National Node of the ESFRI **European RI ELIXIR**, a distributed e-Infrastructure aiming at the construction of a sustainable European infrastructure for biological information. **My role** on the project was to develop containerized-versions of various software tools.
- 2018 - 2020 **RECONNECT** PhD
RECONNECT is an Interreg V-B "Balkan-Mediterranean 2014-2020" project. It aims to develop strategies for sustainable management of Marine Protected Areas (MPAs) and Natura 2000 sites. **My role** was to conduct amplicon analysis for microbial communities.

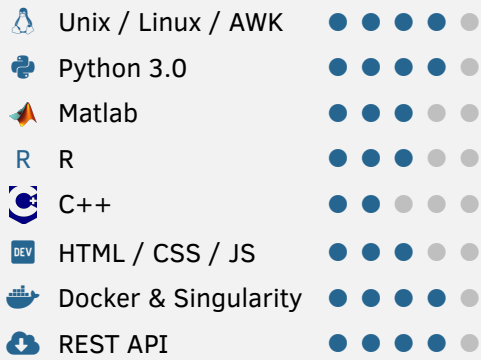
Unofficial Contributions

- 2021 - 2023 **EMBRC** PhD
The European Marine Biological Resource Centre (EMBRC) was established in 2013 to advance fundamental and applied marine biology and ecology research, while promoting sustainable blue economy. Over the last years, I have contributed in tasks of the **Greek node** in the development of workflows for the analysis of the data produced by the network.

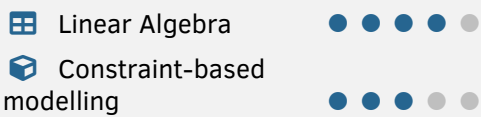
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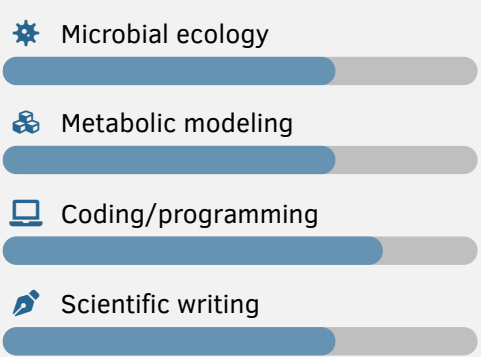
Coding Skills



Math and Stats



Barskills



2020 - ongoing

LifeWatch-ERIC

LifeWatch ERIC is a European Research Infrastructure Consortium providing e-Science research facilities to scientists investigating biodiversity and ecosystem functions and services in order to support society in addressing key planetary challenges. Since PEMA was selected to be part of an [Internal Joint Initiative](#), I have been maintaining the workflow to address the needs of the infrastructure.

PhD

Education

Graduate studies

2018 – 2022	PhD in Bioinformatics	University of Crete, Biology department
	Dissertation	IMBBC - HCMR
	Microbial communities through the lens of high throughput sequencing, data integration and metabolic networks analysis	
2016 – 2018	M.Sc. in Bioinformatics	University of Crete, School of Medicine
	grade: 9.1/10.0	
	Master Thesis	IMBBC - HCMR
	eDNA metabarcoding for biodiversity assessment: Algorithm design and bioinformatics analysis pipeline implementation	

Undergraduate studies

2011 – 2016	B.Sc. in Biology	National & Kapodistrian University of Athens
	grade: 6.2 / 10.0	
	Bachelor Thesis	School of Science, department of Biology
	Morphology, morphometry and anatomy of species of the genus <i>Pseudamnicola</i> in Greece	

Publications

2026	Sampling from the Solution Space and Metabolic Environments of Genome-Scale Metabolic Models <i>Zafeiropoulos, Haris and Rios Garza, Daniel</i> arXiv, DOI: 10.48550/arXiv.2603.29546
2025	microbetag: simplifying microbial network interpretation through annotation, enrichment tests, and metabolic complementarity analysis <i>Zafeiropoulos, Haris, Delopoulos, Ermis Ioannis Michail, Erega, Andi, Schneider, Aline, Geirnaert, Annelies, Morris, John and Faust, Karoline</i> Genome Biology, DOI: 10.1186/s13059-025-03769-2
2025	Establishing the ELIXIR Microbiome Community <i>Finn, R. D., Balech, B., Burgin, J., Chua, P., Corre, E., Cox, C. J., Donati, C., dos Santos, V. M., Fosso, B., Hancock, J., Heil, K. F., Ishaque, N., Kale, V., Kunath, B. J., Médigue, C., Nogueira, T., Pafilis, Evangelos, Pesole, G., Richardson, L., Santamaria, M., Strepis, N., Van Den Bossche, T., Vizcaíno, J. A., Zafeiropoulos, Haris, Willassen, N. P., Pelletier, E. and Batut, B.</i> F1000Research, DOI: 10.12688/f1000research.144515.2
2025	μGrowthDB: querying, visualizing, and sharing microbial growth curve data <i>Radev, Andrey, Casado Gómez-Pallete, Julia, Monsalve Duarte, Sofia, Faust, Karoline and Zafeiropoulos, Haris</i> bioRxiv, DOI: 10.1101/2025.10.13.682118

- 2025 **A long-term ecological research data set from the Marine Genetic Monitoring Program ARMS-MBON 2018–2020**
Daraghmeh, Nauras, Exter, Katrina, Pagnier, Justine, Balazy, Piotr, Cancio, Ibon, Chatzigeorgiou, Giorgos, Chatzinikolaou, Eva, Chelchowski, Maciej, Christmas, Nathan Alexis Mitchell, Comtet, Thierry and others
Molecular Ecology Resources, DOI: [10.1111/1755-0998.14073](https://doi.org/10.1111/1755-0998.14073)
- 2024 **Predicting microbial interactions with approaches based on flux balance analysis: an evaluation**
Joseph, Clémence, Zafeiropoulos, Haris, Bernaerts, Kristel and Faust, Karoline
BMC Bioinformatics, DOI: [10.1186/s12859-024-05651-7](https://doi.org/10.1186/s12859-024-05651-7)
- 2024 **Improving genome-scale metabolic models of incomplete genomes with deep learning**
Boer, Meine D., Melkonian, Chrats, Zafeiropoulos, Haris, Haas, Andreas F., Rios Garza, Daniel and Dutilh, Bas E.
iScience, DOI: [10.1016/j.isci.2024.111349](https://doi.org/10.1016/j.isci.2024.111349)
- 2024 **dingo: a Python package for metabolic flux sampling**
Chalkis, Apostolos, Fisikopoulos, Vissarion, Tsigaridas, Elias and Zafeiropoulos, Haris
Bioinformatics Advances, DOI: [10.1093/bioadv/vbae037](https://doi.org/10.1093/bioadv/vbae037)
- 2024 **Unveiling emerging opportunistic fish pathogens in aquaculture: a comprehensive seasonal study of microbial composition in Mediterranean fish hatcheries**
Skliros, Dimitrios, Kostakou, Maria, Kokkari, Constantina, Tsertou, Maria Ioanna, Pavludi, Christina, Zafeiropoulos, Haris, Katharios, Pantelis and Flietakis, Emmanouil
Microorganisms, DOI: [10.3390/microorganisms12112281](https://doi.org/10.3390/microorganisms12112281)
- 2023 **metaGflow: a workflow for the analysis of marine Genomic Observatories shotgun metagenomics data**
Zafeiropoulos, Haris, Beracochea, Martin, Ninidakis, Stelios, Exter, Katrina, Potirakis, Antonis, De Moro, Gianluca, Richardson, Lorna, Corre, Erwan, Machado, João, Pafilis, Evangelos and others
GigaScience, DOI: [10.1093/gigascience/giad078](https://doi.org/10.1093/gigascience/giad078)
- 2023 **Metabolic models of human gut microbiota: advances and challenges**
Rios Garza, Daniel, Gonze, Didier, Zafeiropoulos, Haris, Liu, Bin and Faust, Karoline
Cell Systems, DOI: [10.1016/j.cels.2022.11.002](https://doi.org/10.1016/j.cels.2022.11.002)
- 2022 **PREGO: a literature and data-mining resource to associate microorganisms, biological processes, and environment types**
Zafeiropoulos, Haris, Paragkamian, Savvas, Ninidakis, Stelios, Pavlopoulos, Georgios A., Jensen, Lars Juhl and Pafilis, Evangelos
Microorganisms, DOI: [10.3390/microorganisms10020293](https://doi.org/10.3390/microorganisms10020293)
- 2022 **Deciphering the community structure and the functional potential of a hypersaline marsh microbial mat community**
Pavludi, Christina and Zafeiropoulos, Haris
FEMS Microbiology Ecology, DOI: [10.1093/femsec/fiac141](https://doi.org/10.1093/femsec/fiac141)
- 2022 **Automating the curation process of historical literature on marine biodiversity using text mining: the DECO workflow**
Paragkamian, Savvas, Sarafidou, Georgia, Mavraki, Dimitra, Pavludi, Christina, Beja, Joana, Eliezer, Menashé, Lipizer, Marina, Boicenco, Laura, Vandepitte, Leen, Perez-Perez, Ruben and others
Frontiers in Marine Science, DOI: [10.3389/fmars.2022.940844](https://doi.org/10.3389/fmars.2022.940844)
- 2021 **0s and 1s in marine molecular research: a regional HPC perspective**
Zafeiropoulos, Haris, Gioti, Anastasia, Ninidakis, Stelios, Potirakis, Antonis, Paragkamian, Savvas, Angelova, Nelina, Antoniou, Aglaia, Danis, Theodoros, Kaitetzidou, Eliza, Kasapidis, Panagiotis and others
GigaScience, DOI: [10.1093/gigascience/giab053](https://doi.org/10.1093/gigascience/giab053)

2021	The Dark mAtteR iNvestigator (DARN) tool: getting to know the known unknowns in COI amplicon data <i>Zafeiropoulos, Haris, Gargan, Laura, Hintikka, Sanni, Pavloudi, Christina and Carlsson, Jens</i> Metabarcoding and Metagenomics, DOI: 10.3897/mbmg.5.69657
2021	Geometric Algorithms for Sampling the Flux Space of Metabolic Networks <i>Chalkis, Apostolos, Fisikopoulos, Vissarion, Tsigaridas, Elias and Zafeiropoulos, Haris</i> 37th International Symposium on Computational Geometry (SoCG 2021), DOI: 10.4230/LIPIcs.SocG.2021.21
2021	The Santorini Volcanic Complex as a valuable source of enzymes for bioenergy <i>Polymenakou, Paraskevi N., Nomikou, Paraskevi, Zafeiropoulos, Haris, Mandalakis, Manolis, Anastasiou, Thekla I., Kiliadis, Stephanos, Kyrpides, Nikos C., Kotoulas, Georgios and Magoulas, Antonios</i> Energies, DOI: 10.3390/en14051414
2020	PEMA: a flexible pipeline for environmental DNA metabarcoding analysis of the 16S/18S ribosomal RNA, ITS, and COI marker genes <i>Zafeiropoulos, Haris, Viet, Ha Quoc, Vasileiadou, Katerina, Potirakis, Antonis, Arvanitidis, Christos, Topalis, Pantelis, Pavloudi, Christina and Pafilis, Evangelos</i> GigaScience, DOI: 10.1093/gigascience/giaa022

Conferences

2025	Poster presentation: \texttt{microbetag}: simplifying microbial network interpretation through annotation, enrichment and metabolic complementarity analysis 24th Annual International Conference on Systems Biology (ICSB 2025) @ Dublin, Ireland	ICSB25
2024	Poster presentation: \texttt{microbetag}: simplifying microbial network interpretation through annotation, enrichment and metabolic complementarity analysis 19th International Symposium on Microbial Ecology @ Cape Town, South Africa	ISME19
2024	Poster presentation: \texttt{microbetag}: simplifying microbial network interpretation through annotation, enrichment and metabolic complementarity analysis Belgian Society for Microbiology: Annual Symposium @ Brussels, Belgium	BSM2024
2022	Poster presentation: \texttt{microbetag}: a microbial co-occurrence network annotator 1st Applied HoloGenomics conference @ Bilbao, Spain	AHC2022
2022	Poster presentation: Reverse ecology and systems biology approaches to identify key processes ensuring life in extreme environments 18th International Symposium on Microbial Ecology @ Lausanne, Switzerland	ISME18
2021	Flash talk: \texttt{dingo}: A python library for metabolic networks sampling & analysis Bioinformatics Open Source Conference (BOSC) - online	BOSC2021
2021	Flash talk: PEMA v2: addressing metabarcoding bioinformatics analysis challenges 1st DNAQUA International Conference - online	DNAQUA2021
2020	Flash talk: Mining literature and -omics (meta)data to associate microorganisms, biological processes and environment types Federation of European Microbiological Societies (FEMS) - online	FEMS2020

2020	Oral presentation: Geometric and statistical methods in systems biology: the case of metabolic networks PyData Global - online	PyData2020
2019	Participation network2019: 4th Symposium on Ecological Networks @ Paris, France	Network2019
2019	ePoster presentation: P.E.M.A.: a Pipeline for Environmental DNA Metabarcoding Analysis 8th International Barcode of Life Conference @ Trondheim, Norway	iBOL2019
2018	Poster presentation: P.E.M.A.: a Pipeline for Environmental DNA Metabarcoding Analysis European Conference on Computational Biology (ECCB) 2018 @ Athens, Greece	ECCB18

Workshops & Summer schools

2025	School of artificial intelligence applied to microbiomes @ AgroParisTech, France - Sampling the solution space of metabolic networks (theory & hands-on)	SAIM
2025	Data integration with Microbial Networks and Community Models @ KULeuven, Belgium - <i>Organizing committee</i> - microbetag: metabolic secrets behind microbial co-occurrence (theory & hands-on)	DIMNCM
2025	Metabolic models applied to microbiomes @ Antony, France - <i>Organizing committee</i> - Metabolic model analysis methods (theory & hands-on)	MMAM
2023	Summer School on Microbial Community Design and Control @ Leuven, Belgium Assisting on "metabolic network reconstruction" topic	DCMC
2023	Economic Principles in Cell Physiology @ Paris, France Mathematical modeling of cellular systems and "resource allocation thinking". <i>Poster presentation:</i> "saml'em all": exploring potential flux sampling applications	EPCP2023
2022	Microbial communities: current approaches and open challenges @ Cambridge, UK Interdisciplinary exchanges in microbial communities' research and the emerging challenge areas.	UMCW06
2022	ELIXIR Fluxomics Training School 2021 An introduction to the field of fluxomics and the experimental and computational methods used to estimate and predict metabolic fluxes	ELIXFLUX
2020	Metagenomics, Metatranscript- omics and multi 'omics for microbial community studies A Physalia Course from Prof. Dr. Curtis Huttenhower	material

Supervision

2025–ongoing	Sourav Pattanaik, MSc student "Populating μ GgrowseDB"	
2024–ongoing	Sotiris Touloupoulos, ERASMUS student "Pre- and post-sampling features to leverage flux sampling at both the strain and the community level"	GitHub repo

2024–ongoing	Xi Peng, Visiting PhD candidate "Enhancing microbetag's functional annotations with quorum sensing prediction capabilities"
2024–ongoing	Anna Voukouna, MSc student "Investigation of microbial metabolic interactions potential across the Genome Taxonomy Database representative genomes"
2024–2025	Andrey Radev, MSc student "Visualisation and analysis of microbial growth data to facilitate interactions investigation and interactions' strength assessment"
2024	Sotiris Touloupoulos, GSoC student blogpost "Pre- and post-sampling features to leverage flux sampling at both the strain and the community level"
2023–2024	Sofia Monsalve Duarte, MSc student "Creation and implementation of a database interface for the collection and analysis of microbial growth data"
2023–2024	Ermis Ioannis Michail Delopoulos, GSoC student post "Development of a Cytoscape App for microbe-microbe association networks"
2022–2024	Ermis Ioannis Michail Delopoulos, MSc student "Development of a graphical user interface for microbetag"
2022–2023	Julia Casado Gómez-Pallete, MSc student thesis "Development of a database for human gut bacterial growth curves"

Teaching

2025	Exploring the microbiome with high-throughput sequencing technologies: a bioinformatics perspective slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace , Greece
2025	Introduction in metabolic modeling GitHub repo Theoretical and practical session on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace , Greece
2024	Exploring the microbiome with high-throughput sequencing technologies: a bioinformatics perspective (part II) slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace , Greece
2024	Exploring the microbiome with high-throughput sequencing technologies: a bioinformatics perspective slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace , Greece
2023	eDNA metabarcoding, pipeline development & high performance computing slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace , Greece
2023	eDNA metabarcoding, pipeline development & high performance computing slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace , Greece
2022	Genome-scale model reconstruction GColab notebook Lectures on "Master en bioinformatique et modélisation" of the Université Libre de Bruxelles

Zafeiropoulos Haris

Bioinformatics, Systems
Biology, Microbial Ecology

Software

pema

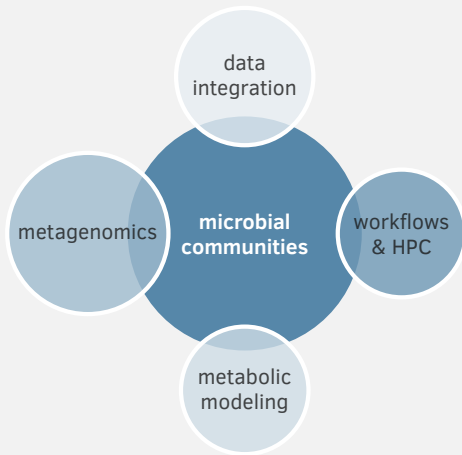
prego

darn

dingo

microbetag

Research fields



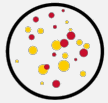
Memberships



Mikrobiokosmos



International Society of Microbial Ecology



Belgian Society for Microbiology

2021

Microbiome Data Analyses Workshop Online

MDAWO

Tutorial: PEMA: a flexible Pipeline for Environmental DNA Metabarcoding Analysis of the 16S/18S rRNA, ITS and COI marker genes.

2020

Environmental DNA & DNA metabarcoding

slides

Lecture on MSc "Environmental Biology" in [University of Crete](#)

2019

8th Conference of Hellenic Society of Mikrobiokosmos

MKB8

Tutorial: PEMA: a flexible Pipeline for Environmental DNA Metabarcoding Analysis of the 16S/18S rRNA, ITS and COI marker genes.

Awards & Scholarships

2021

9th conference of the Hellenic Society of Mikrobiokosmos

MBK9

Oral presentation award: Sampling the flux space of microbial metabolic networks for (un)biased assessment of the possible functional states: the example of Sars-Cov-2 on the human alveolar macrophage metabolic network.

2021

Google Summer of Code

GSoC

Project title: *From DNA sequences to metabolic interactions: building a pipeline to extract key metabolic processes*

2021

European Molecular Biology Organization Short-Term Fellowship

EMBO

Project title: *"Exploiting data integration, text-mining and computational geometry to enhance microbial interactions inference from co-occurrence networks"*

2020

Federation of European Microbiological Societies Meeting Attendance Grant

FEMS

for joining the *"Metagenomics, Metatranscript-omics and multi 'omics for microbial community studies"* Physalia course

2019

Short Term Scientific Mission (STSM) @ DNAqua-net COST action

DNAqua-net

Project title *A comparison of bioinformatic pipelines and sampling techniques to enable benchmarking of DNA metabarcoding.* [Report](#).

2018

Best Poster Award @ Hellenic Bioinformatics conference

HBIO

for *PEMA: a Pipeline for Environmental DNA Metabarcoding Analysis*

References

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